

Practice №1. Forecasting on a base of a multiple linear regression model

1. Find a data set by selecting one dependent variable and 7 or 8 independent variables (factors) that have, in your opinion, the influence on the dependent variable. The sample must contain at least 50 values;
2. For this data set you need to conduct a statistical analysis for each variable by calculating its numerical characteristics: mean variance, mode, median, kurtosis, asymmetry, coefficient of variation, range and others;
3. Construct a correlation matrix and analyze closeness of the relationship between the dependent variable and each factor;
4. Build a multiple linear regression model that includes all available factors.
5. Give interpretation for all coefficients.
6. Evaluate the quality and reliability of the constructed regression equation: calculate standard errors of the equation and regression coefficients. Highlight significant and insignificant coefficient estimates (significant level $\alpha = 0,05$). Construct confidence intervals for the model coefficients.
7. Calculate the coefficient of determination and the adjusted coefficient of determination. Interpret them. What is the difference between them?
8. Check if there is the multicollinearity of factors in the model.
9. Construct a multiple linear regression model using forward method of including factors into the model based on the positive growth of the adjusted coefficient of determination.
10. Verify statistical significance of all regression coefficients and the equation as a whole at the 5% level of significance; interpret the obtained estimates of the model coefficients.
11. Check the model residuals for systematic errors using an appropriate test.
12. Verify graphically if the condition of homoskedasticity for the residuals is fulfilled.
13. By using an appropriate test verify if the condition of homoskedasticity for the residuals is fulfilled.
14. Check for the absence of autocorrelation by graphically.
15. Check the assumption of independence between residuals using the Durbin-Watson and Breusch–Godfrey tests.
16. Check if the residuals are normally distributed using one of specified tests.
17. In case of failure of the Gauss-Markov conditions, try to correct undesirable effects by transforming variables, or by removing outliers, or by conducting the new regression specification and identification, etc.
18. Generate point and interval forecasts of the dependent variable given the factors growth by 10% and 20% of their mean. If some factor represents a dummy variable the forecasts for all possible cases should be given.